

Maxime Lafleur

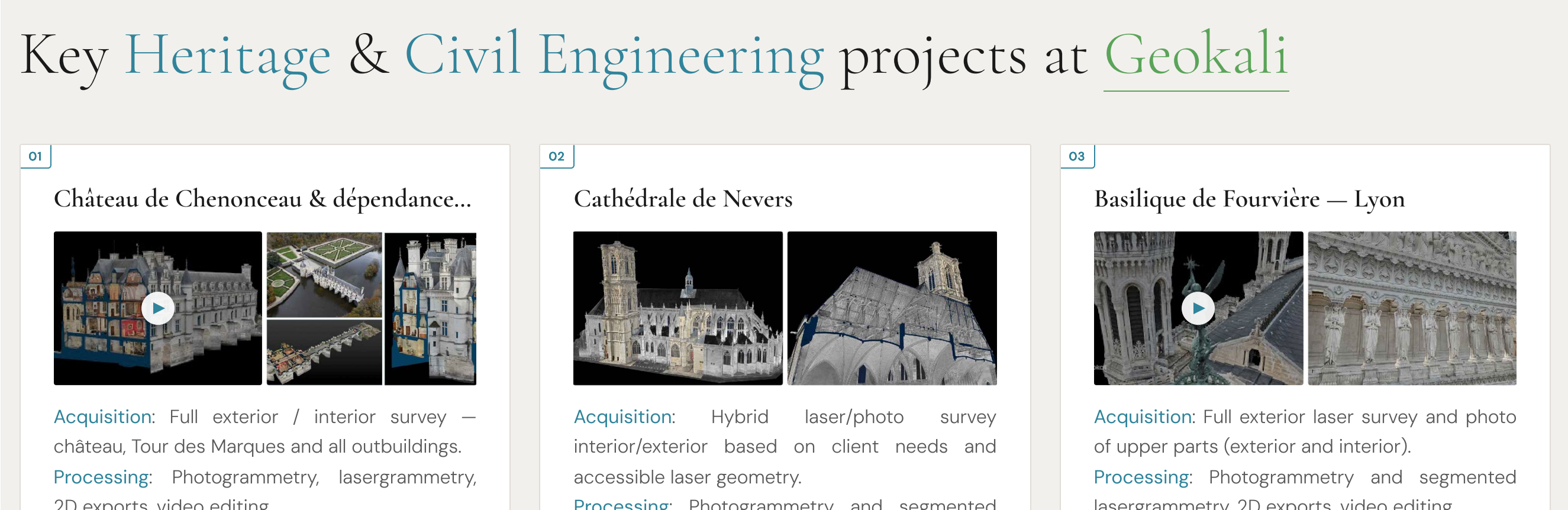
Engineer - Photogrammetrist - Lasergrammetrist - Developer

Passionate about cultural heritage and cutting-edge 3D reconstruction technologies, I initially built my expertise around drone geomatics — from hardware and software development to applied research — gaining full autonomy over the entire process, from study and prototyping to product development (2014–2022).

More recently (2022–2026), my work in the cultural heritage sector immersed me in high-fidelity 3D capture and reconstruction. The technical challenges and the blending of diverse technologies meant mastering specialized skills and methodologies, pushing existing solutions to their limits, and developing automation to eliminate repetitive tasks.

With 3D technology evolving rapidly and opening up new possibilities, I'm eager to keep exploring this field, learn from others and draw on new approaches, as part of a creative team — bringing my 12 years of curiosity and varied experience.

Looking forward to connecting and discussing projects, opportunities and experiences!

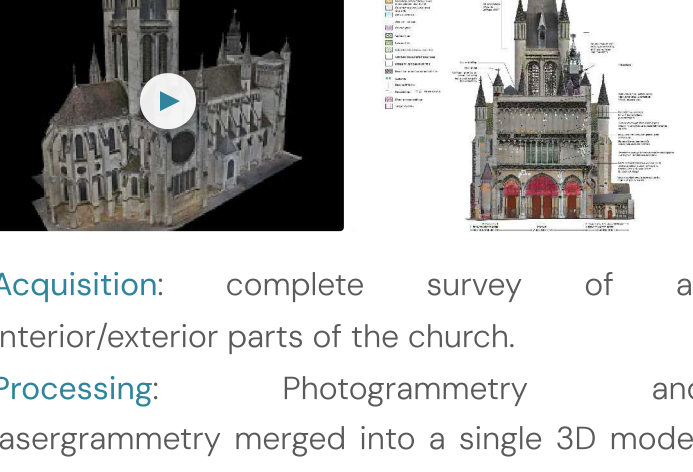


— GEOKALI - FIELD PROJECTS

Key Heritage & Civil Engineering projects at Geokali

01

Château de Chenonceau & dépendance...



Acquisition: Full exterior / interior survey — château, Tour des Marques and all outbuildings.

Processing: Photogrammetry and lasergrammetry merged into a single 3D model, 2D exports, video editing.

Acquisition: 3 months / 2 years

Processing: 3 months

Images: ~285 000

3D Models: ~150 - 193M Δ

Project time: 16 TB

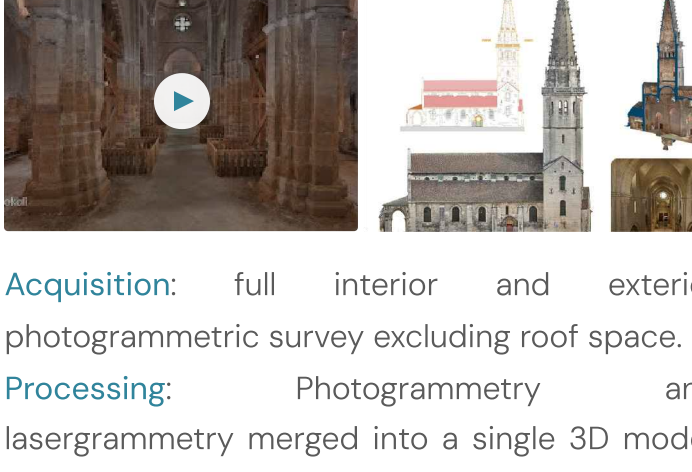
Scenes: ~5 500

Texture resolution: 1mm

2D AutoCAD drawings: 173 .dwg - ~700 ortho

02

Cathédrale de Nevers



Acquisition: Hybrid laser/photo survey interior/exterior based on client needs and accessible laser geometry.

Processing: Photogrammetry and segmented lasergrammetry, 2D exports and video editing.

Acquisition: 1 month

Processing: 2 months

Images: ~40 000

3D Models: ~30 - 237M Δ

2D AutoCAD drawings: 57 .dwg - ~200 ortho

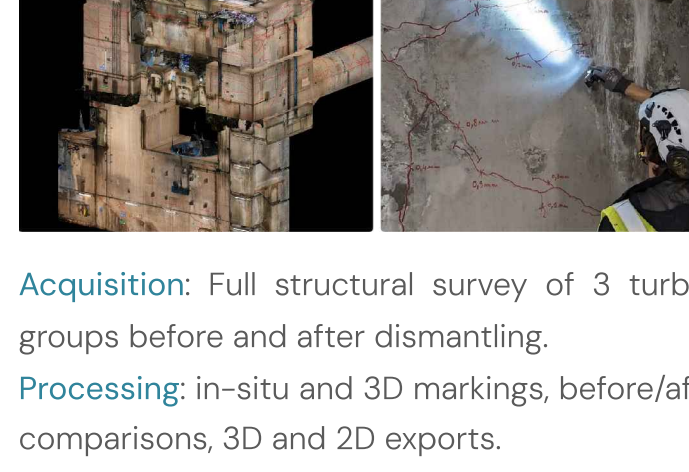
Scenes: ~1 830

Texture resolution: 1mm

2D AutoCAD drawings: 57 .dwg - ~200 ortho

03

Basilique de Fourvière — Lyon



Acquisition: Full exterior laser survey and photo of upper parts (exterior and interior).

Processing: Photogrammetry and segmented lasergrammetry, 2D exports, video editing.

Acquisition: 2 weeks

Processing: 1 month

Images: ~48 000

3D Models: ~20 - 189M Δ

Project time: 57 .dwg - ~200 ortho

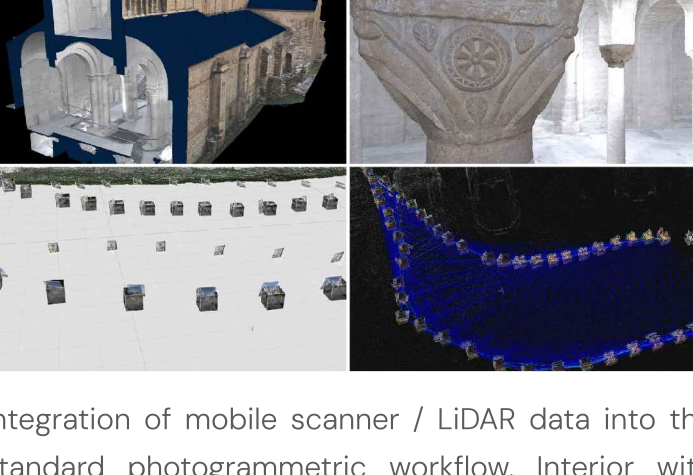
Scenes: ~400

Texture resolution: 1mm

2D AutoCAD drawings: 57 .dwg - ~200 ortho

04

Église Notre-Dame — Dijon



Acquisition: complete survey of all interior/exterior parts of the church.

Processing: Photogrammetry and lasergrammetry merged into a single 3D model, 2D exports with variants, video editing.

Acquisition: 3 weeks

Processing: 2 months

Images: ~54 000

3D Models: ~2 - 140M Δ

2D AutoCAD drawings: 40 .dwg - ~160 ortho

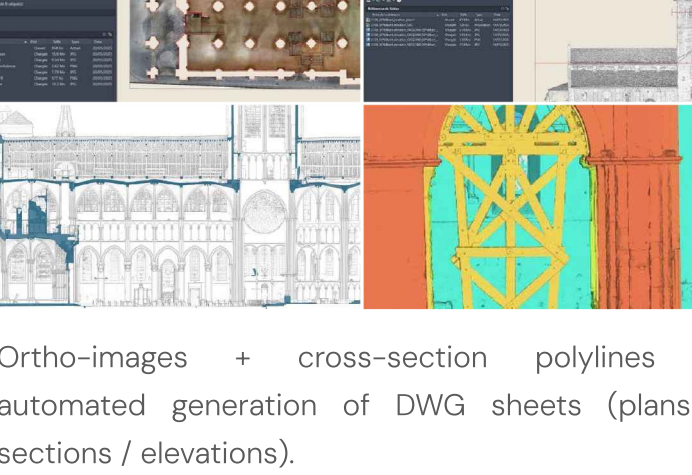
Scenes: ~1 100

Texture resolution: 1mm

2D AutoCAD drawings: 40 .dwg - ~160 ortho

05

Église Saint-Philibert — Dijon



Acquisition: full interior and exterior photogrammetric survey excluding roof space.

Processing: Photogrammetry and lasergrammetry merged into a single 3D model, 2D exports, video editing.

Acquisition: 2 weeks

Processing: 1 month

Images: ~30 000

3D Models: ~30 - 19M Δ

2D AutoCAD drawings: 4 .dwg - ~40 ortho

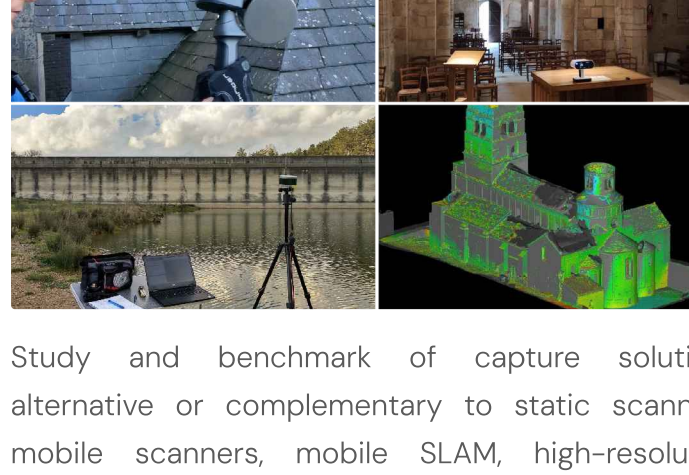
Scenes: ~250

Texture resolution: 1mm

2D AutoCAD drawings: 4 .dwg - ~40 ortho

06

Civil Engineering Structure



Acquisition: Full structural survey of 3 turbine groups before and after dismantling.

Processing: in-situ and 3D markings, before/after comparisons, 3D and 2D exports.

Acquisition: 2+ 2 weeks

Processing: 9 months

Images: ~60 000

3D Models: 3 - 50M Δ

Project time: 8 TB

Scenes: ~400

Texture resolution: 0.35 mm

Defects (Optima 3DS): ~900 m

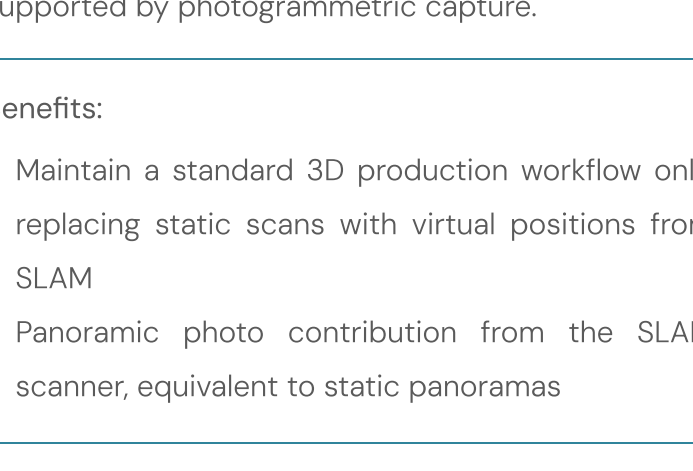
2D AutoCAD drawings: 30 .dwg - ~120 ortho

— INNOVATION & DEVELOPMENT

Key Heritage R&D projects at Geokali

01

Mobile LiDAR data integrated into the photogrammetric pipeline



Integration of mobile scanner / LiDAR data into the standard photogrammetric workflow. Interior with mobile scanner, exterior with mobile scanner supported by photogrammetric capture.

Benefits:

- Replacing a standard 3D production workflow only, maintain static scans with virtual positions from SLAM
- Panoramic photo contribution from the SLAM scanner, equivalent to static panoramas

Costs:

- Georeferencing precision, drift analysis, strict GCP setup — strong mastery of SLAM solutions and their limits
- Risk of geometric fidelity failure (drift, double surfaces) with no correction capacity during processing

Study site: Abbaye de Cluny

Ground plane acquisition: 2 hours

Texture resolution (in-ground photogrammetric resolution): 1mm

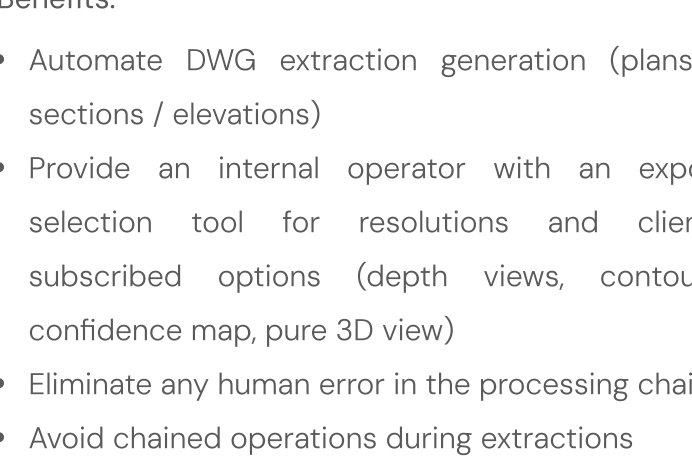
SLAM acquisition: 2+ 45 min along trajectory

Scenes: 1/2 m

Texture resolution (in-ground photogrammetric resolution): 1mm

02

Automated 2D DWG sheet generation



Ortho-images + cross-section polylines — automated generation of DWG sheets (plans / sections / elevations).

Benefits:

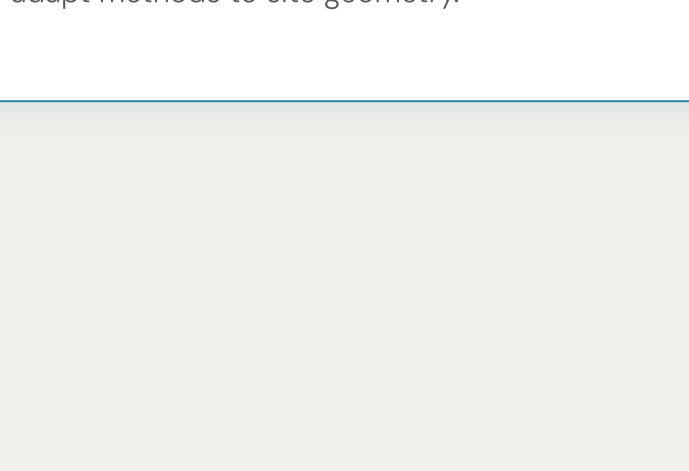
- Automate DWG extraction generation (plans / sections / elevations)
- Provide an internal operator with an export selection tool for resolutions and client-subscribed options (depth views, contours, confidence map, pure 3D view)
- Eliminate any human error in the processing chain
- Avoid chained operations during extractions
- Systematize deliverables and lock nomenclature, extraction themes and renders
- Respond quickly to client requests — extraction productivity multiplied by >30 per .dwg sheet

Costs:

- SDK software development, trial and error with trusted partners, maintenance across new software versions.

03

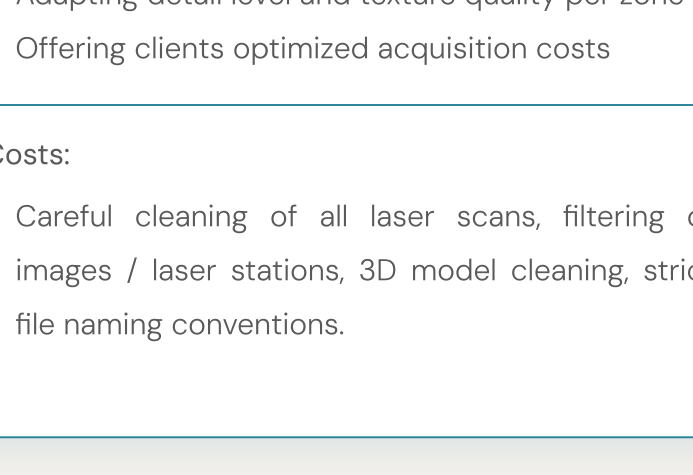
Technology watch



Study and benchmark of capture solutions alternative or complementary to static scanners: mobile scanners, mobile SLAM, high-resolution photogrammetry — to optimize acquisition time and adapt methods to site geometry.

04

3D Production by segmentation



Benefits:

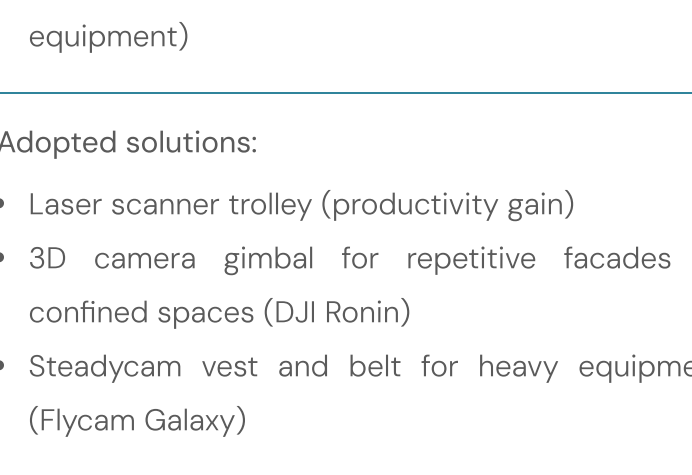
- Reducing 3D processing time and complexity
- Splitting work between 3D operators and computers
- Ability to reprocess a specific area following potential changes
- Adapting detail level and texture quality per zone
- Offering clients optimized acquisition costs

Costs:

- Careful cleaning of all laser scans, filtering of images / laser stations, 3D model cleaning, strict file naming conventions.

05

Hardware prototyping



Optimizing acquisition systems by adding purpose-built accessories adapted to field constraints.

Identified needs:

- Accessing confined spaces
- Securing the system for specific use cases
- Simplifying daily operations (heavy handheld equipment)

Adopted solutions:

- Laser scanner trolley (productivity gain)
- 3D camera gimbal for repetitive facades or confined spaces (DJI Ronin)
- Steadycam vest and belt for heavy equipment (Flycam Galaxy)

Custom solutions:

- Low-profile tripod for terrestrial scanner (daily use)
- Camera addon fixed to terrestrial laser scanner
- Custom attachment systems for gimbals

06

Software prototyping



Pushing daily-use software to its limits to improve working comfort, deliverable quality and processing time, and to develop key algorithms for acquisition / processing methodologies.

Automated script development for:

- AutoCAD / Photoshop / Leica Cyclone 3DR / Agisoft Metashape / RealityScan

Developed tools:

- Multi-scale analysis tool for alignment discontinuities from RealityScan photogrammetric processing
- Automated image QC tool (motion/focal blur, exposure) with renaming, copying and raw data backup
- Ortho-image generation with various renders: depth, contours, Xray, 3D confidence, geometric voids — to streamline the drafter's workflow

Small algorithms:

- Image filtering by location and orientation
- WGS84 to local projected coordinates conversion
- Storage redundancy detection script
- Automated conversion of panoramic to equirectangular images
- Injection of pre-calibrated camera calibrations into EXIF metadata
- And other small useful developments on a project-by-project basis

— PROFESSIONAL EXPERIENCE

Geokali

2022 — JUN 2026 - 4 YRS

Microdrones™ / GeoCue

2016 — 2022 - 6 YRS 5 MO

CHRS

2021 — 2022 - 1 YR 6 MO

Delair

2016 — 2018 - 2 YRS

INVENTUAV

2015 — 2016 - 1 YR

Vision du Ciel Intl

2014 — 2016 - 1 YR

Groupe SII - Nexter

2013 — 2014 - 1 YR

Photogrammetry / Lasergrammetry Engineer

Geokali - Full-time - Valence, France

3 yrs

Oct. 2022 — present

3D data acquisition and processing on **cultural heritage** and **civil engineering** sites, using a full sensor fleet: UAV drones, terrestrial laser scanners (TLS), mobile laser scanners (MLS), DSC cameras and structured light lasers.

→ [See Heritage & Civil Engineering projects](#) · [See R&D projects](#)

Jean-Marc Anneron — Draftsman-Surveyor-Laser scanning specialist, Historic monuments since 2001

“ I had the pleasure of working alongside Maxime Lafleur for several years on demanding operations related to surveying and graphical restitution of historic monuments. Looking back (and as a Heritage draftsman with over 25 years of experience), I can say without hesitation that he is among the best specialists in the field I have had the chance to collaborate with. Maxime has an extremely rare profile, combining technical rigidity, adaptability, creativity and rigor. Where many see a blocker or a constraint, he already starts designing a solution he studies, develops routines, automates complex tasks and creates efficient methodologies that push 3D/2D tool outputs to their maximum. The quality, readability and ease of use of his deliverables reflect this. In the field, his mastery of capture operations is remarkable. Whether in 3D digitization (photogrammetry, laser scanning), he shows great rigor and an excellent reading of situations, which perfectly matches and feeds Geokali's expertise. But beyond the skills, what I probably appreciate most about him is the human solidity he brings to a collaboration. Maxime is approachable, reliable, attentive, proactive and generously involved. When working with him, you immediately feel you can move forward with confidence, even on the most complex projects.”

PHOTOGRAMMETRY

LIDAR-TLS/MLS

UAV

3D RECONSTRUCTION

LEICA SDK

DWG AUTOMATION

POINT CLOUDS

HARDWARE PROTOTYPING

CULTURAL HERITAGE

R&D

UAV LiDAR Engineer

Microdrones™ / GeoCue - R&D - Toulouse

6 yrs

May 2016 — Sept. 2022

Within the R&D department of Microdrones / GeoCue, responsible for developing **new UAV-embedded LiDAR prototypes**, designing high-precision LiDAR targets, and QC of new processing software.

— Operational setup of Microdrones France subsidiary (office design, construction follow-up, HR)

— Hardware design of UAV LiDAR systems

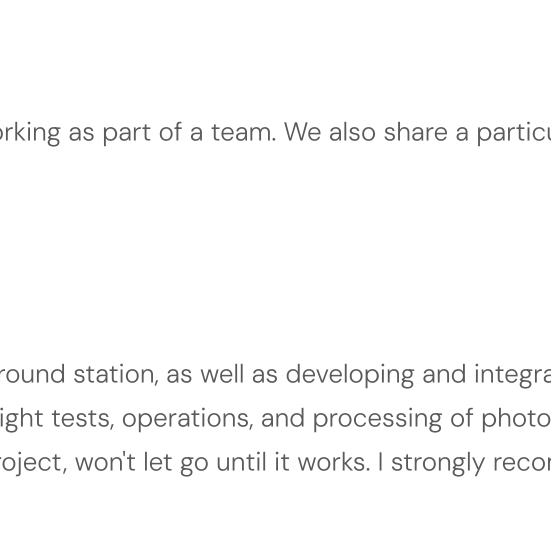
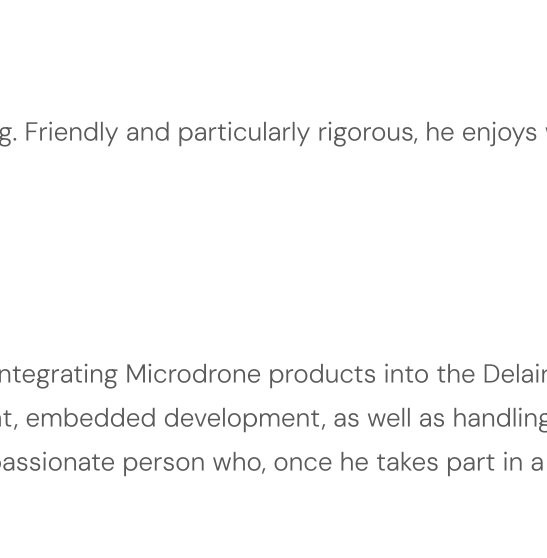
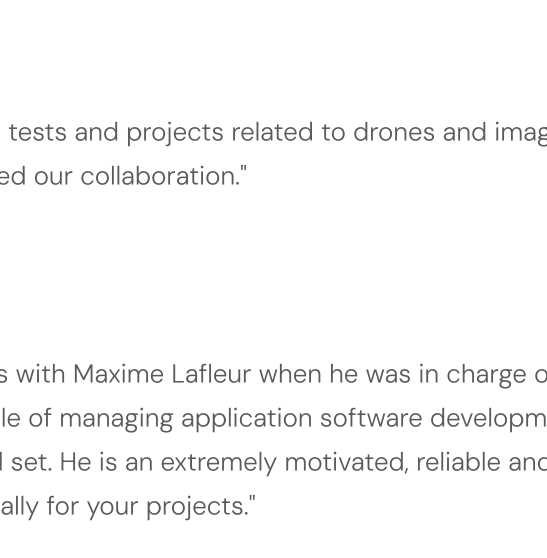
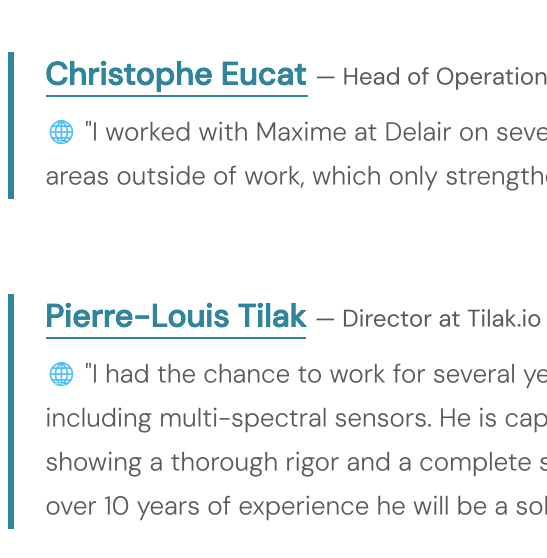
— Field testing and data QC

— Hardware and software benchmarks

— Development of an automated tree counting tool

— Automated powerline classification

— Parachute system development and EASA certification approval



Vincent Eucatat

— Industrial CAD Draftsman at midGroup

“ I had the opportunity to work alongside Maxime Lafleur during my time at Microdrones, and I can wholeheartedly vouch for his excellent skills as an engineer. As part of our drone data acquisition missions for various vineyard estates in France, Maxime was a true technical pillar. His analytical mindset and great resourcefulness allowed him to efficiently diagnose and solve complex hardware and software issues, often under demanding field conditions. Whether it was last-minute adjustments to the equipment or optimizing flight operations, his ingenuity always ensured the continuity and quality of our operations. I highly recommend him to any organization looking for a reliable and talented field engineer.”

Erwan Gavelle

— Founder of geopid

“ I had the pleasure of working with Maxime on the development of drone payloads. He was in charge of the electronics and project management while I worked on the mechanics. It was my first job in this field, but he trusted me, taught me and challenged me throughout the projects. I particularly appreciated his skills, his involvement and his friendliness in driving the projects to success. He also managed a large part of the organization and setup of our offices in France. I highly recommend Maxime for his professionalism, his good mood and his ability to support the people he works with.”

UAV LiDAR R&D

HARDWARE DESIGN

EASA CERTIFICATION

POWERLINE CLASSIFICATION

TREE COUNTING

FIELD TESTING

3D / GIS Support for Antarctic PhD Project

CNRS - Civic Service - Remote

1.5 yrs

Feb. 2021 — Jul. 2022

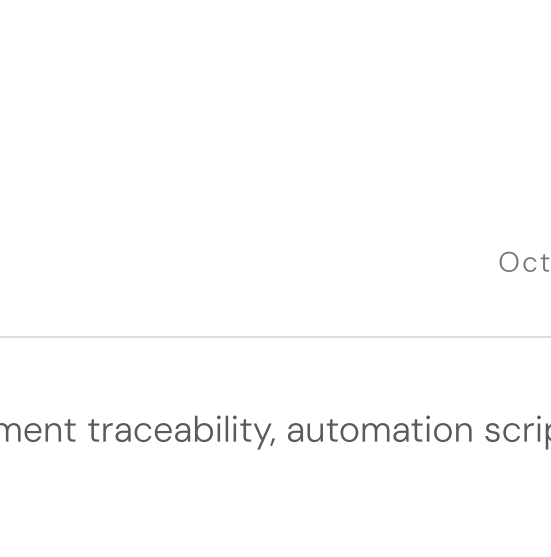
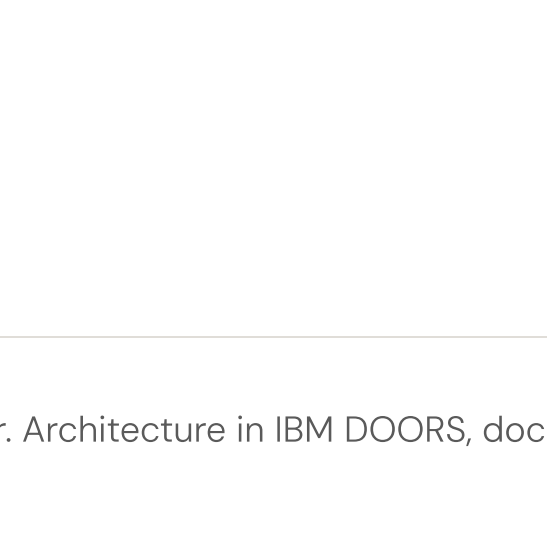
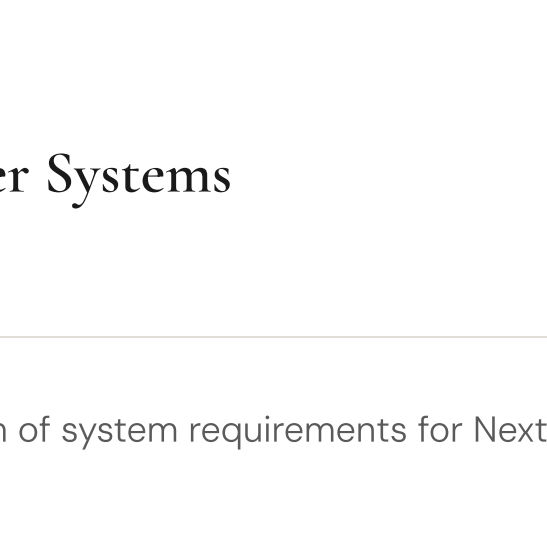
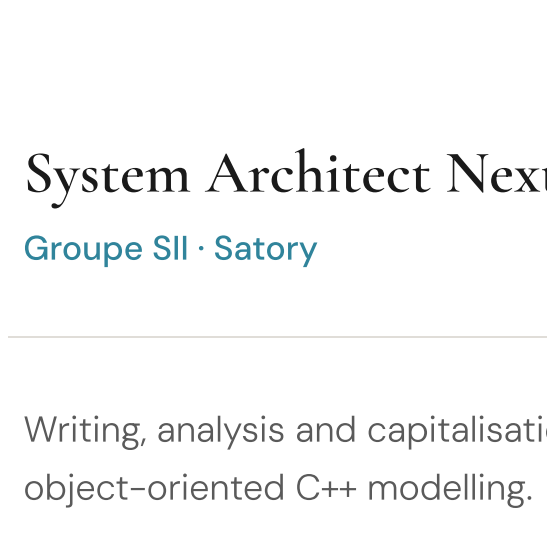
Technical and methodological support for a **PhD project** on the **spatial distribution of penguins in Antarctica**. 3D processing of 11 Antarctic sites.

— GNSS base station PPP post-processing

— PPK LiDAR processing

— Multi-site photogrammetry

— High-precision Ortho-DEM generation



ANTARCTICA

PPK LiDAR

GNSS PPP

PHOTOGRAMMETRY

ORTHO-DEM

GIS

UAV Engineer Payload Integration & Control/Command

Delair - Full-time - Toulouse

2 yrs

May 2016 — May 2018

Within Delair-Tech's R&D department, under Microdrones' strategic direction, responsible for:

— Control/command system development for Microdrones UAVs

— Integration of Delair-Tech payloads (multispectral, optronic, photogrammetry) on Microdrones md4-1000

— Viticulture research project in partnership with Naïo Technologies (vineyard row mapping with PPK)

— Other prototyping projects

Christophe Eucatat

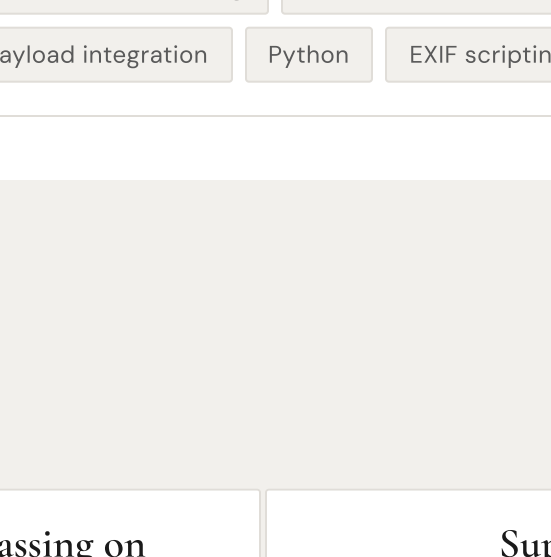
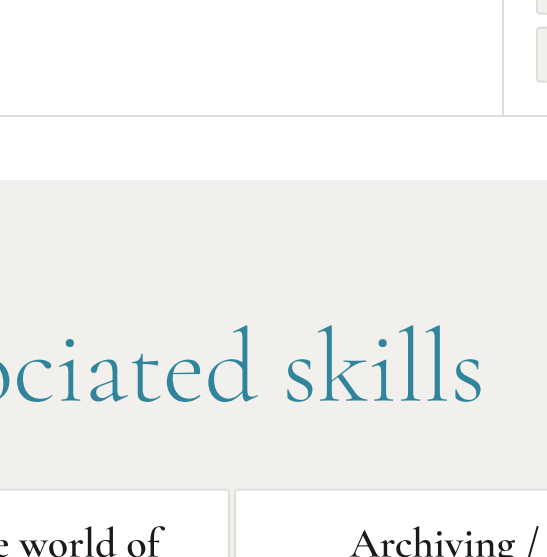
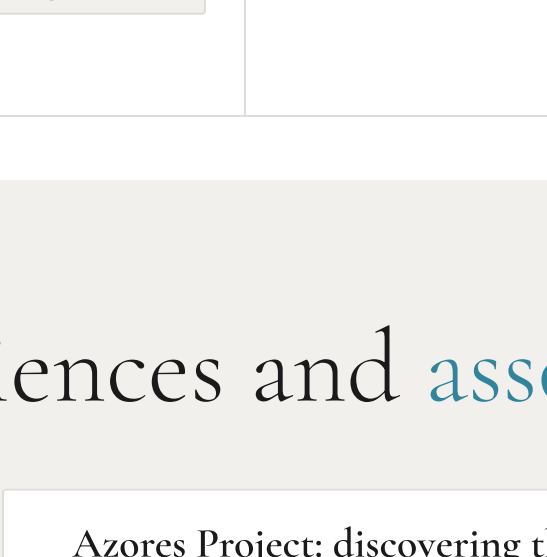
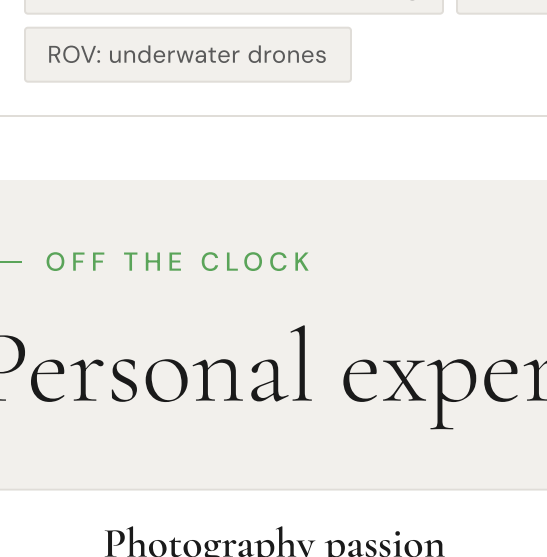
— Head of Operations

“ I had the chance to work on Delair on several tests and projects related to drones and imaging. Friendly and particularly rigorous, he enjoys working as part of a team. We also share a particular interest in several areas outside of work, which only strengthened our collaboration.”

Pierre-Louis Tiliket

— Director at Tialcio

“ I had the chance to work for several years with Maxime Lafleur when he was in charge of integrating Microdrones products into the Delair ground station, as well as developing and integrating several payloads, including multi-spectral sensors. He is capable of managing application software development, embedded development, as well as handling flight tests, operations, and processing of photogrammetric data, showing a thorough grasp and a complete skill set. He is an extremely motivated, reliable and passionate person who, once he takes part in a project, won't let go until it works. I strongly recommend him, as with over 10 years of experience he will be a solid ally for your projects.”



PAYLOAD INTEGRATION

FIXED-WING UAV

EMBEDDED SYSTEMS

Co-Founder

InventUAV - Open-Source Startup - France

1 yr

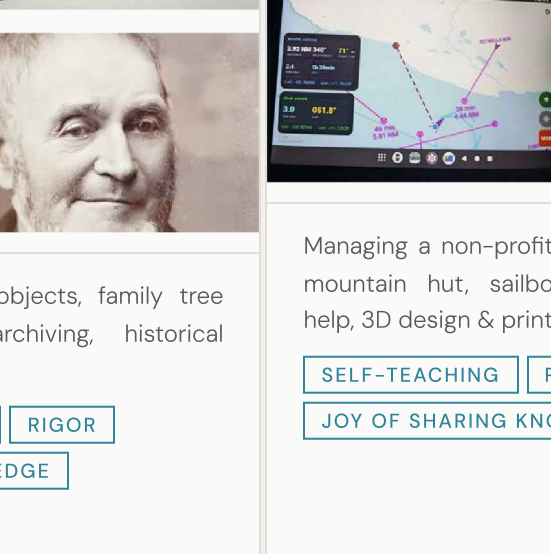
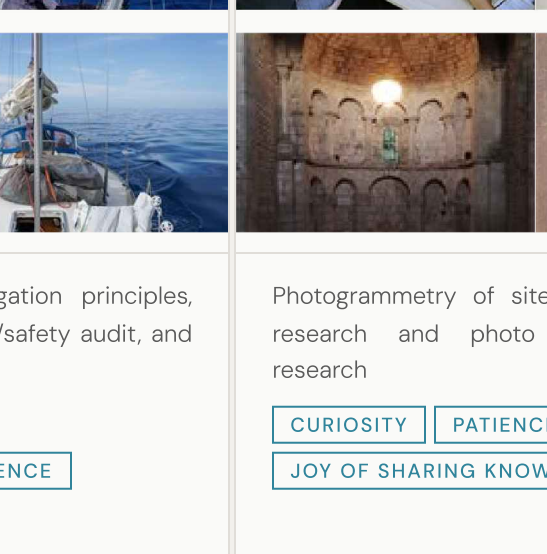
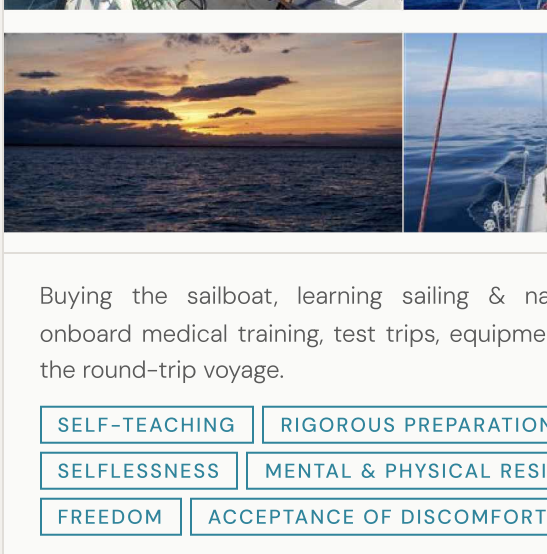
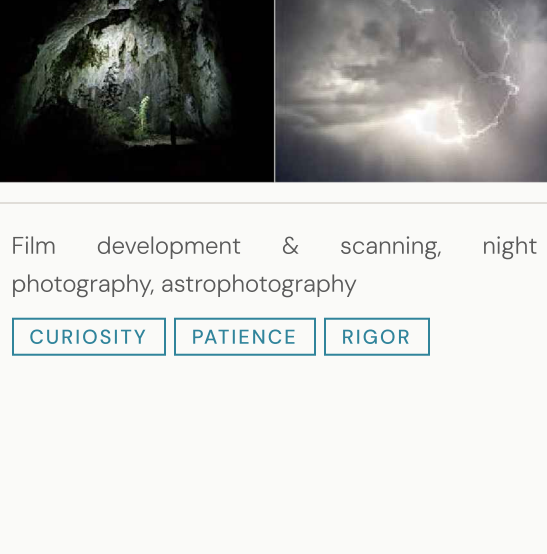
May 2015 — May 2016

Co-founder of an open-source UAV startup developing an **autonomous indoor drone** for warehouse management and inventory.

— ArduPlane plugin for Gazebo / ROS

— ArduCopter debugging

— Visual recognition for indoor missions



ARDUPILOT

ROS / GAZEBO

OPEN SOURCE

INDOOR DRONE

UAV Apps Engineer (GCS / GUI)

Vision du Ciel Intl - Orange

1 yr

Jul. 2014 — Jun. 2015

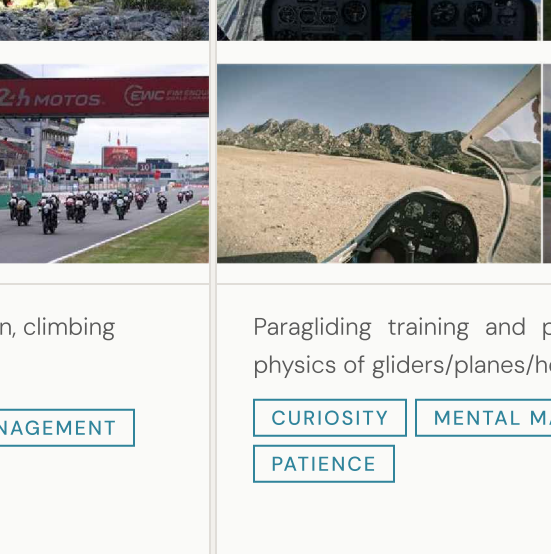
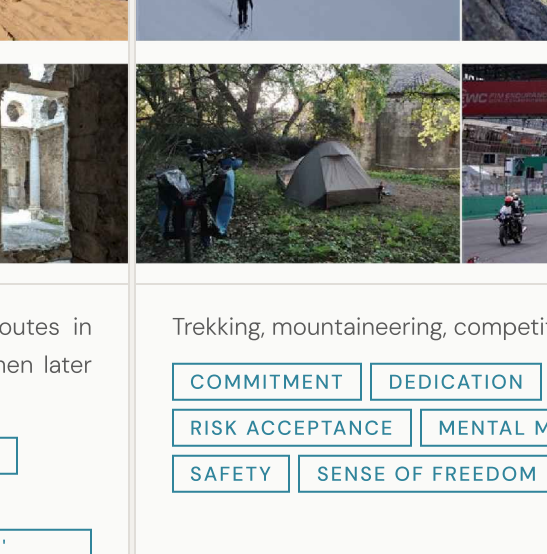
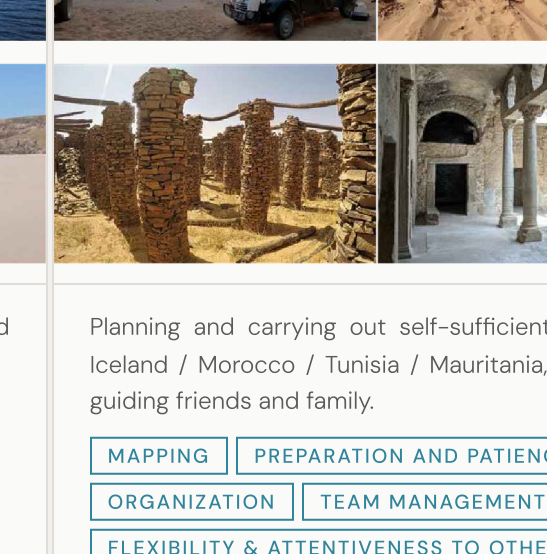
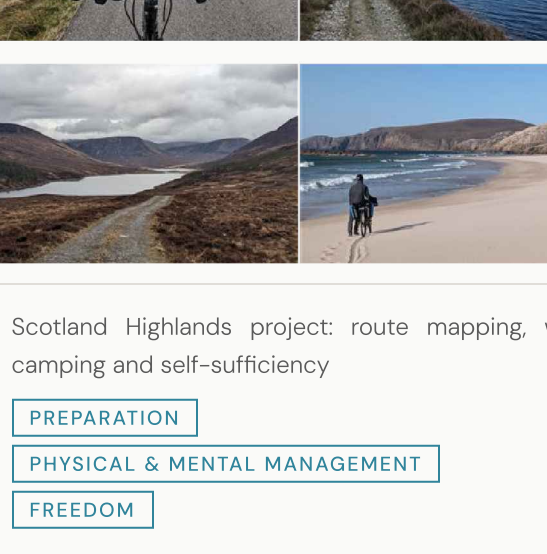
Development of **UAV mission control** software solutions on Windows: mission planning, control, supervision and reporting.

— GUI development in C#, JS, Python

— Embedded development in C++

— Open-source and industrial protocol integration

— Prototyping: UAVs, radio controllers, gimbals, parachutes



GCS DEVELOPMENT

C# / PYTHON / JS

PIXHAWK

HARDWARE PROTOTYPING

System Architect Nexter Systems

Groupe SII - Satory

1 yr

Oct. 2013 — Jul. 2014

Writing, analysis and capitalisation of system requirements for Nexter. Architecture in IBM DOORS, document traceability, automation script development, object-oriented C++ modelling.

SYSTEM ARCHITECTURE

IBM DOORS

C++

REQUIREMENTS ANALYSIS

More information available on request

— TECHNICAL EXPERTISE

Skills & tools (non exhaustive)

Acquisition & Field

Hardware prototyping (tripod, steadycam, gimbal)

UAV / drone (multirotor, fixed-wing)

Camera calibration (intrinsic & extrinsic)

Aerial & terrestrial photogrammetry

MLS — Mobile Laser Scanning and Lidars

Field quality control (QA/QC)

GNSS / PPK / PPP processing

TLS — Terrestrial Laser Scanning

RDV underwater drones

Software

RealityCapture / RealityScan / PostShot / Metashape

Point cloud processing & cleaning

GIS (QGIS)

2+F Laser Control Office / Scenaria

Unreal Engine 5 / Cinema 4D

Solidworks & 3D printing

Postshot & 3DGS

AutoCAD

RIGOROUS PREPARATION

SELF-TEACHING

SELFLESSNESS

FREEDOM

COMMITMENT

RISK ACCEPTANCE

SAFETY

JOY OF SHARING KNOWLEDGE

DEDICATION

MENTAL MANAGEMENT

SENSE OF FREEDOM

ACCEPTANCE OF DISCOMFORT

Development

Multi-source 3D reconstruction (photo + TLS + MLS + structured light)

Automated tools (plan extraction, QC, AI denoising)

AI experiments (photogrammetry, Android apps)

System architecture & requirements (LMI, DOORS)

ArduPilot / APM

Scanner hardware benchmarking

GUI development (Windows, GCS mission control)

Electronic prototyping

Payload integration

Python

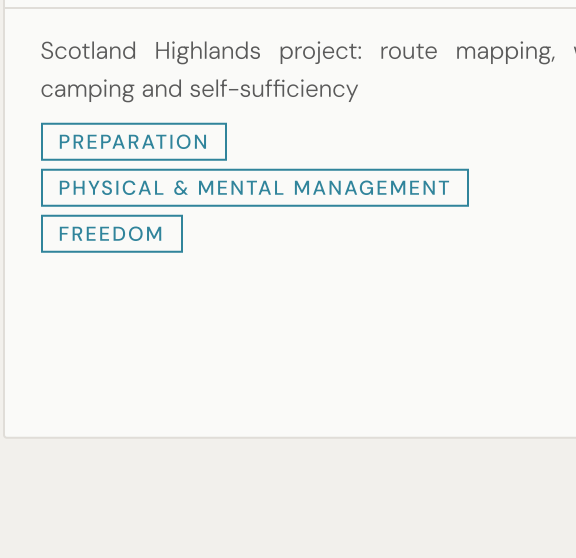
EXIF scripting

C++ / C#

— OFF THE CLOCK

Personal experiences and associated skills

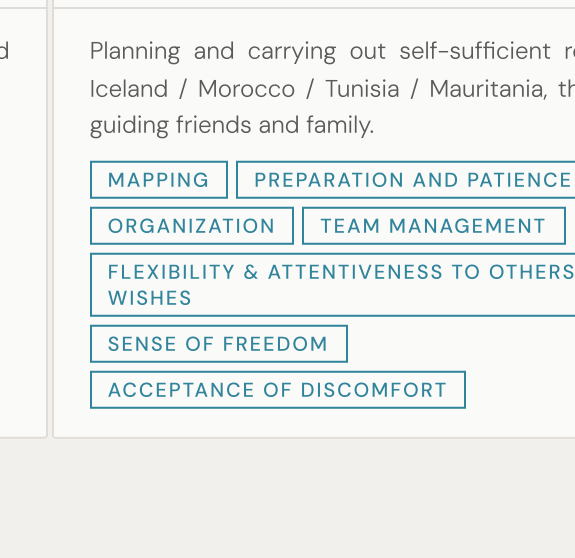
Photography passion



Film photography & scanning, night development, self-sufficiency

CURIOSITY | PATIENCE | RIGOR

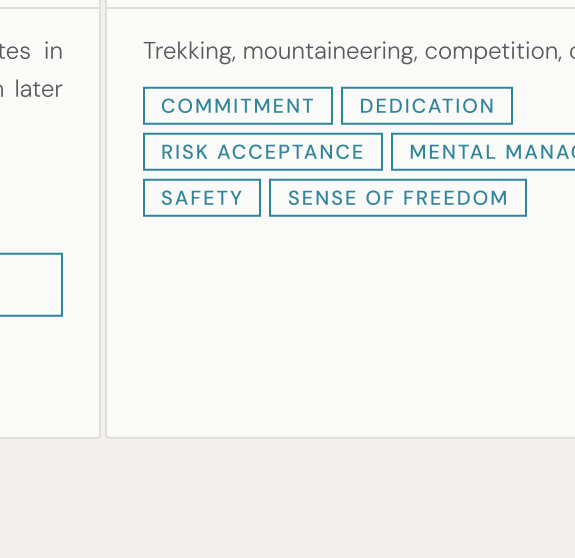
Azores Project: discovering the world of sailing



Buying the sailboat, learning sailing & navigation principles, onboard medical training, test trips, equipment/safety audit, and the north-rip voyage.

SELF-TEACHING | RIGOROUS PREPARATION | SELFLESSNESS | MENTAL / PHYSICAL RESILIENCE | FREEDOM

Archiving / passing on knowledge

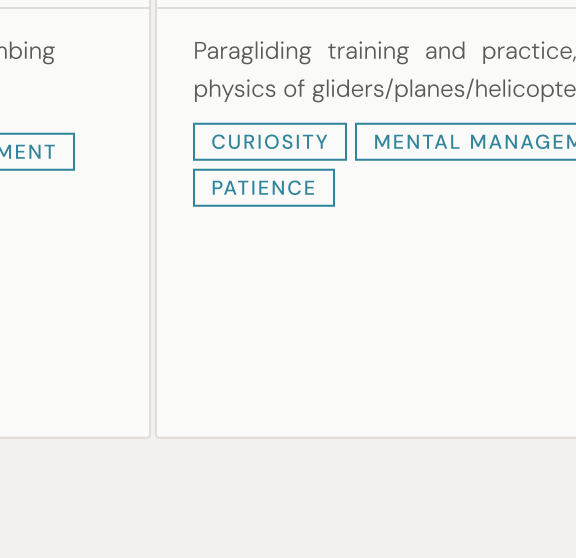


Photogrammetry of stereosubjects, family tree research and photo archiving, historical research

CURIOSITY | PATIENCE | RIGOR

JOY OF SHARING KNOWLEDGE

Support

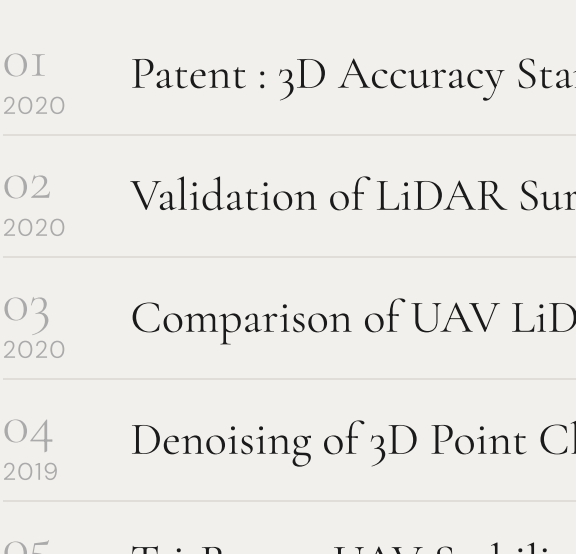


Managing a non-profit website, opening a CAF mountain hut, sailboat delivery, mechanical help, 3D design & printing, Android apps

SELF-TEACHING | PATIENCE

JOY OF SHARING KNOWLEDGE

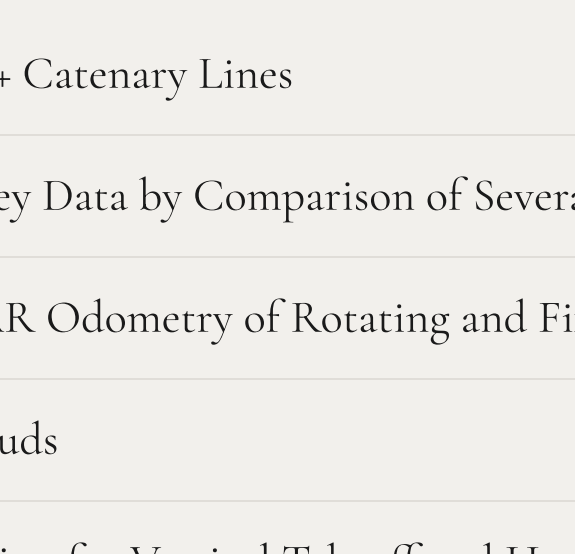
Highlands Project: cycling



Scotland Highlands project: route mapping, wild camping and self-sufficiency

PREPARATION | ORGANIZATION | TEAM MANAGEMENT | FREEDOM

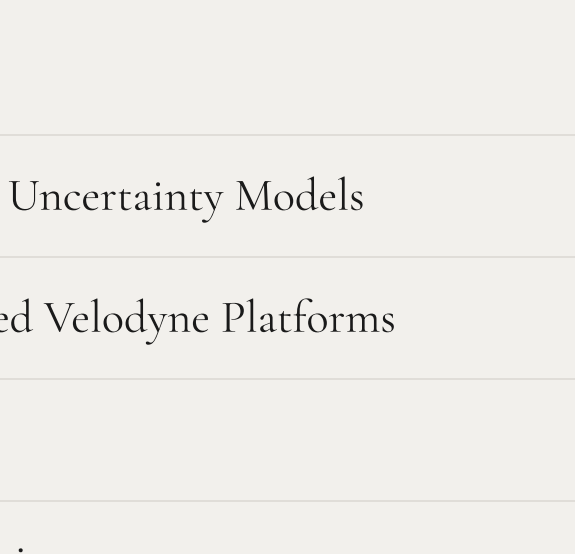
Desert Discovery



Planning and carrying out self-sufficient routes in Iceland / Morocco / Tunisia / Mauritania then later guiding friends and family

MAPING | PREPARATION AND PATIENCE | ORGANIZATION | TEAM MANAGEMENT | FLEXIBILITY & ATTENTIVENESS TO OTHERS | SENSE OF FREEDOM | ACCEPTANCE OF DISCOMFORT

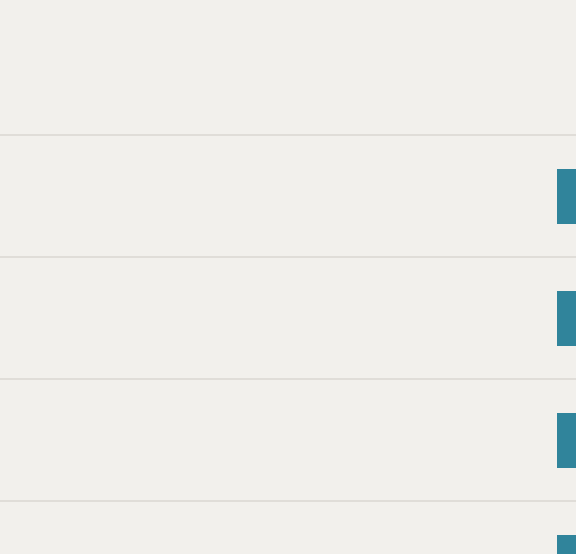
Active pursuits



Trekking, mountaineering, competition climbing

COMMITMENT | DEDICATION | RISK ACCEPTANCE | MENTAL MANAGEMENT | SAFETY | SENSE OF FREEDOM

Aviation passion



Paragliding training and practice, discovering the physics of gliders/planes/helicopters, simulators

CURIOSITY | MENTAL MANAGEMENT | RIGOR

PATIENCE

— RESEARCH & INNOVATION

Publications & Patents

01 2020

Patent : 3D Accuracy Star + Catenary Lines

PATENT

02 2020

Validation of LiDAR Survey Data by Comparison of Several Uncertainty Models

PUBLICATION

03 2020

Comparison of UAV LiDAR Odometry of Rotating and Fixed Velodyne Platforms

PUBLICATION

04 2019

Denoising of 3D Point Clouds

PUBLICATION

05 2022

Tri-Rotors UAV Stabilization for Vertical Takeoff and Hovering

PUBLICATION

Education & International exchanges

EU

European Drone Certificate

Open AI/A2/A3 and Specific CAFE

2026

EEC

Master of Engineering (MEng) Electrical & Electronics Engineering

2008–2013

US

University of California, Irvine

Exchange Program Global Management & Manufacturing

2012

CA

Concordia University

Semester Exchange Engineering & Computer Science

2011

— CONTACT

Let's connect !

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LINKEDIN

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3D SPACE SKETCHFAB

maxxiuv

EMAIL

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MOBILITY

Rhône-Alpes → France → Europe → World

LANGUAGES

French (native) - English (Full Professional)

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